Basics of Microscope Assembly

Project: Group Management (INFIBIO)

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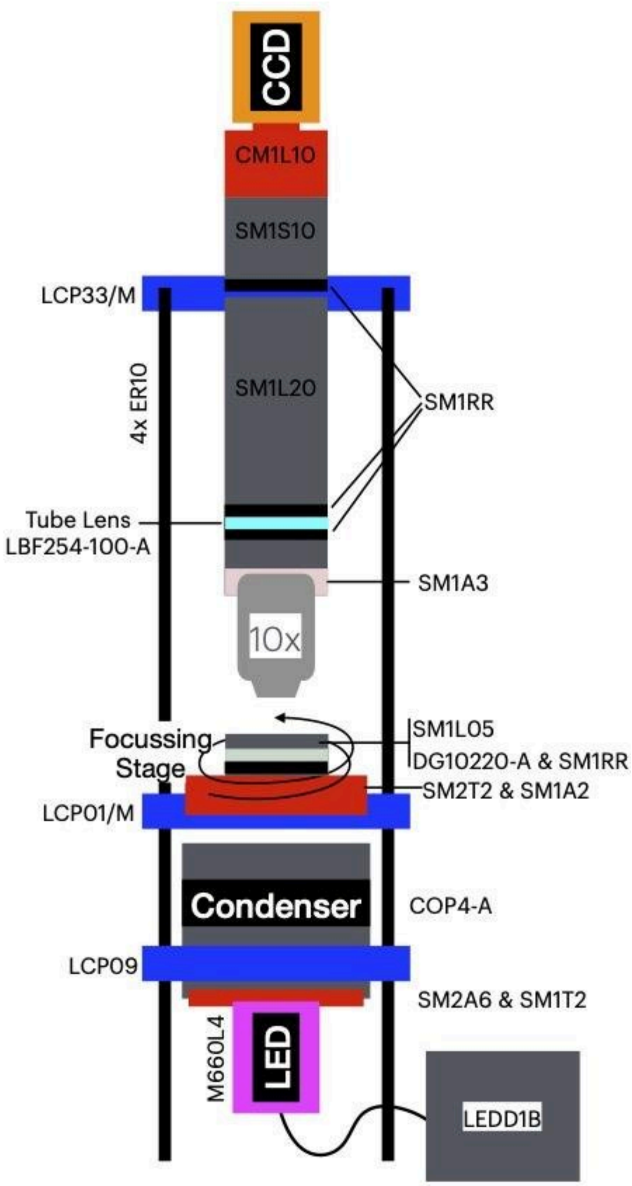
For light-driven motility in microalgae, we will use a simple, portable microscope built with Thorlabs components, a 10X microscope objective, a CCD camera, and an LED.

Schematics of the setup:

ICTP_microscope_schematics.jpg



Description of the setup

A microscope objective is held on a lens tube system together with a tube lens and a CCD camera. The tube lens projects the image of the focal plane of the objective onto the CCD camera (which means that the CCD camera is at a focal distance away from the tube lens). The tube is held on a cage system which includes a simple stage in front of the objective, and a illumination system behind the stage. The separation between the stage and the objecive can be changed by rotating the stage. The illumination is realised through a condenser lens collimating as well as possible the light from a LED. A frosted glass is inserted just below the place where the sample is located, in order to improve the uniformity of the illumination field.

A Note on the objective-tube lens choice

The objective is a RMS-threaded ***infinity-corrected* objective.** Despite being in principle infinity-corrected, for optimal performance these lenses should still be used with the appropriate tube lens and separation between objective and tube lens. Here we are not using the correct tube lens and we are not adjusting the separation between the lens and the back of the objective. This introduces potential aberrations in the image that would otherwise have been corrected (to the best possible) by using the correct tube lens and correct separation. **This is fine for us because we are still using a low magnification and low NA.** However, it would be visible with high-end objectives, which would then form images of lower quality due to aberrations. If you want to know more about the Infinity Optical System, you can start here. Tube lenses can be purchased e.g. from Edmund Optics (see here for Zeiss objectives).

Part list

Thorlabs

Note: the LED condenser can be replaced with a SM2 tube lens and an appropriate aspheric condenser, perhaps already with a diffuser surface. The illumination LED and its cube driver can be replaced with a cheaper LED and driver, possibly from Adafruit or RS. This should reduce significantly the cost.

Thorlabs parts

	Part number	Description	Unit Cost	Quantity	Total Cost
1	ER10	cage assembly rod	12.73	4	50.92
2	CM1L10	Extension tube	20.76	1	20.76
3	SM1S10	Lens tube spacer	13.29	1	13.29
4	SM1L05	SM1 lens tube 0.5"	12.38	1	12.38
5	SM1L20	SM1 lens tube 2"	16.23	1	16.23
6	SM1T2	SM1 coupler	21.49	1	21.49
7	SM1A2	SM2-SM1 Adapter	25.82	1	25.82
8	SM1A3	SM1-RMS Adapter for RMS-threaded objectives	18.02	1	18.02
9	SM1RR	SM1 retaining rings	4.43	4	17.72
10	SM2A6	SM2-SM1 adapter	26.45	1	26.45
11	SM2T2	SM2 coupler	37.53	1	37.53
12	LCP01/M -->> LCP34/M	60mm cage plate, SM2 threaded	41.04	1	41.04
13	LCP09-->> LCP36	60mm cage plate for SM2 lens tubes	45.61	1	45.61
14	LCP33/M	SM1-Threaded 30 mm to 60 mm Cage Plate Adapter, 0.5" Thick	42.11	1	42.11
15	LBF254-100-A	N-BK7 best form lens, 1"diam, f=100mm, AR coating 350-700nm	56.97	1	56.97
16	DG10-220-A	N-BK7 200grit diffuser, 1" diam, AR coating 350-700nm	46.97	1	46.97

17	COP4-A	Zeiss Collimator adapter, AR coating 350-700nm	199.76	1	199.76
18	M660L4	660nm, mounted LED	233.24	1	233.24
19	LEDD1B	T-Cube LED Driver	322.86	1	322.86
20					
21	**Total**				1249.17

Camera

- IDS UI-3370CP-NIR-GL

Camera specs can be found here and in this file

UI-3370CP-NIR-GL_Rev_2.pdf

This camera uses uEye software, which can be downloaded here. Notice that there is a Python module to interface with uEye cameras, which you can find here.

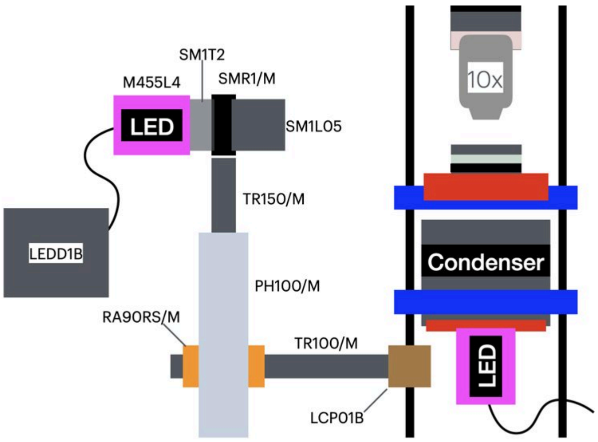
Note: the camera could be replaced e.g. with a Raspberry Pi camera. This should reduce the costs significantly.
Potential cameras to consider include:

- C-mount HQ Camera. 12Mp with C-mount and no front lens. It should be useable as-is as microscope camera and apparently it has a good quality sensor.
- Camera Module 3 Raspberry Pi. This has an autofocus on board and a front lens. The front lens needs to be removed for this to be used as camera for the microscope.

Light-stimulation add-on

The microscope comes with a light-stimulation add-on to study the phototactic response of the cells in the sample. Here is the schematics with the codes of the Thorlabs pieces:

Screenshot 2024-11-15 at 17.14.01.png



Note: 1) Costs might be cut by using a different light source; 2) it might be convenient to include a 1" diam frosted aspheric condenser to the SM1 tube in front of the LED, in order to collimate the LED light as much as possible. To this end one would place the lens with the planar side facing the LED source, and adjust its position so that the LED is at the back focal plane of the lens. The lens can be retained in place with SM1 retaining rings.

Interesting Microscopy Links

- FIJI: software for image manipulation
- MyScope Microscopy Training
- Nikon MicroscopyU
- Edmund Optics Knowledge Centre